THE ISLAMIC UNIVERSITY
College of Technical Engineering
Department of Computer Techniques Engineering



LAB 01 · WEEK 1

Introduction to Cisco Packet Tracer and IOS Configuration Review

Build a small two-router network in Packet Tracer, choose the right cables, and apply and check a basic Cisco IOS configuration.

Packet Tracer window and modes	Choosing the right cable	IOS command modes and CLI help
Basic router and switch configuration	Verification with show and ping	Simulation mode: ARP and ICMP

Course	Computer Networks Protocols	**Code**	CET4102 · 6 ECTS
Stage	Fourth stage · Semester 7	**Session**	Laboratory session · Week 1
Instructor	Asst. Lect. Laith H. Alzubaidi	**Academic year**	2026 – 2027

Prepared for the fourth-stage students of the Department of Computer Techniques Engineering. Primary text: J. F. Kurose & K. W. Ross, *Computer Networking: A Top-Down Approach*, 7th ed. Recommended: D. E. Comer, *Internetworking with TCP/IP*.

LEARNING OUTCOMES

By the end of this lab you will be able to:

1. build a network in Packet Tracer and switch between Realtime and Simulation mode;
2. choose the correct cable for each pair of devices;
3. give IP settings to the PCs and move between the Cisco IOS command modes;
4. apply and save a basic configuration on a router and a switch;
5. check the result with `show` commands and `ping`, and follow a ping in Simulation mode.

L1.1 Background

L1.1.1 Cisco Packet Tracer

Cisco Packet Tracer (PT) is a free program from the Cisco Networking Academy that imitates a network on your computer. It is a *simulator*: it copies the behaviour of routers, switches and PCs in software, so it accepts most, but not all, real Cisco commands. PT has two modes. In **Realtime mode** the network runs all the time, like real equipment. In **Simulation mode** time moves only when you click. Every packet becomes an event that you can step through and open layer by layer (the layers of Lecture 1).

L1.1.2 Choosing the right cable

An Ethernet port sends on one pair of wires and receives on another. PCs and routers send on pins 1 and 2 and receive on pins 3 and 6; this port type is called **MDI**. Switches and hubs do the opposite; their ports are **MDI-X**. The sending pair at one end must reach the receiving pair at the other end. This gives one simple rule:

- **different** port types (PC–switch, router–switch): **straight-through** cable;
- **same** port type (PC–PC, switch–switch, router–router, and also **PC–router**): **cross-over** cable.

The figure below shows this rule as a table (left) and the pin wiring of each copper cable (right).

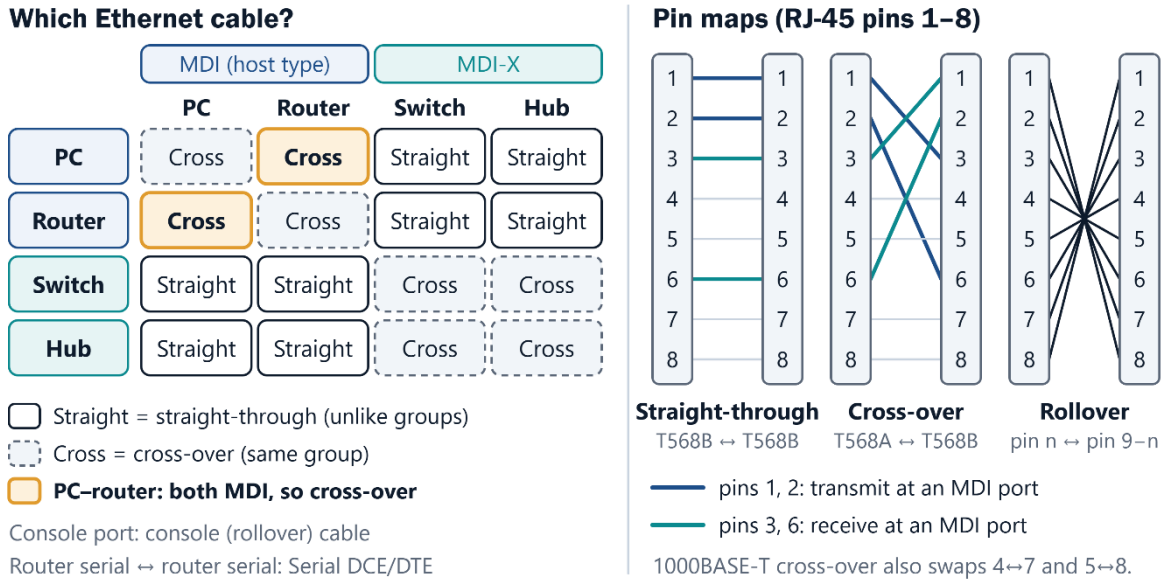


Figure L1.1 Choosing an Ethernet cable (left) and the wiring of the three copper cables (right).

> **COMMON MISTAKE**
>
> Many modern ports have **Auto-MDIX** and correct a wrong cable by themselves. Do not rely on it: in PT, a PC–router link with a straight-through cable stays down.

Two more cables are needed. The **console cable** (a rollover cable) joins a PC's serial (RS-232) port to the **Console** port of a router or switch. It carries only command-line text, so it works before the device has an IP address.

The **serial** cable joins two routers over a WAN link. One end, the **DCE** (data circuit-terminating equipment), supplies the clock that times the bits. The other end, the **DTE** (data terminal equipment), follows that clock.

L1.2 Topology and addressing

You will build the network below: two LANs joined by a serial WAN link between routers R1 and R2. The line style of each link shows the cable to use.

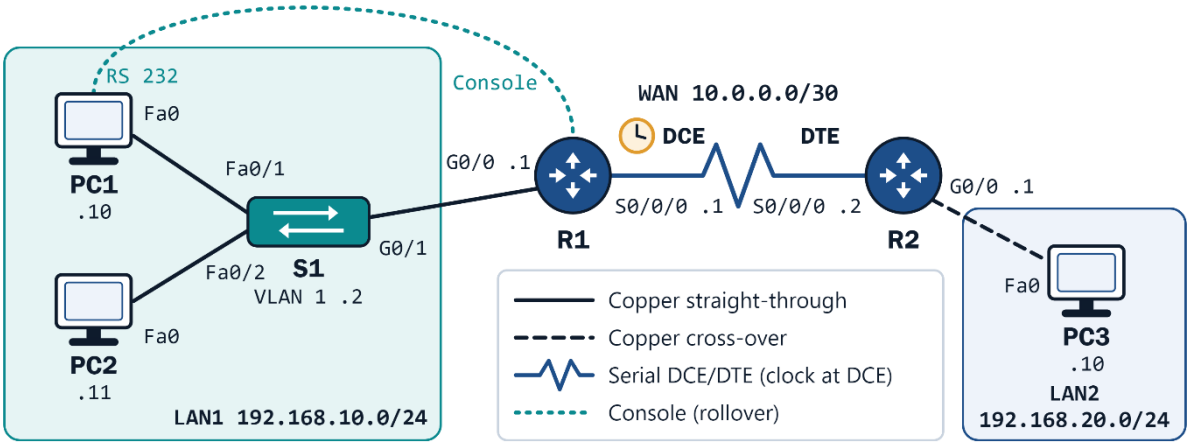


Figure L1.2 Lab topology and cables (".1" = last number of the address, e.g. R1 G0/0 = 192.168.10.1).

All addresses are private (RFC 1918). The WAN mask 255.255.255.252 (/30) leaves exactly two usable addresses, 10.0.0.1 and 10.0.0.2, one for each router; 10.0.0.3 is the broadcast address.

Table L1.1 Addressing table.

Device	Interface	IP address	Subnet mask	Default gateway
R1	G0/0	192.168.10.1	255.255.255.0	N/A
R1	S0/0/0 (DCE)	10.0.0.1	255.255.255.252	N/A
R2	G0/0	192.168.20.1	255.255.255.0	N/A
R2	S0/0/0 (DTE)	10.0.0.2	255.255.255.252	N/A
S1	Vlan1	192.168.10.2	255.255.255.0	192.168.10.1
PC1	Fa0	192.168.10.10	255.255.255.0	192.168.10.1
PC2	Fa0	192.168.10.11	255.255.255.0	192.168.10.1
PC3	Fa0	192.168.20.10	255.255.255.0	192.168.20.1

BEFORE YOU START

- Install Packet Tracer 8.x and log in once with your Networking Academy account.
- For 192.168.10.0/24 and 10.0.0.0/30, write the network address, the first and last usable host and the broadcast address.
- Without looking at the legend of the topology, decide which cable each link needs.

L1.3 Procedure

Save your work as `LAB01_<StudentID>.pkt` after every step.

L1.3.1 Step 1: Build and cable the network

1. Start PT. The large white area is the **Logical** workspace. The device boxes are at the bottom left: pick a device type first, then a model. The tools above the workspace (Select, Delete, Add Simple PDU …) act on devices. The **Realtime** and **Simulation** tabs are at the bottom right.
2. Drag two **1941** routers, one **2960** switch and three **PCs** into the workspace. Label them R1, R2, S1 and PC1–PC3. The label is only a drawing name; the IOS name is set in Step 5.
3. A 1941 has two Gigabit Ethernet ports, `G0/0` and `G0/1`, but no serial port. Click each router, open its **Physical** tab, switch the power off, drag a **HWIC-2T** card into slot 0 and switch the power on. The card adds `Serial0/0/0` and `Serial0/0/1`. (If they appear as `Serial0/1/0` and `0/1/1`, the card is in slot 1: move it.)
4. Connect the devices as in the topology. For the serial link choose **Serial DCE** and click **R1 first**; a clock symbol marks the DCE end. For the console link click PC1 (RS 232), then R1 (Console).
5. Look at the link lights: green = up, red = down, amber = a switch port that is still starting (about 30 s; **Fast Forward Time** skips the wait). The Ethernet and serial links to the routers stay **red**. This is normal: router interfaces are **shut down** until you configure them in Steps 5 and 6.

L1.3.2 Step 2: Configure the PCs

A PC needs an IP address, a subnet mask and a **default gateway**: the router address it uses to reach other networks. On each PC open **Desktop > IP Configuration**, keep **Static** and type the values of the addressing table. Do not ping yet.

L1.3.3 Step 3: Watch a ping in Simulation mode

The network is new, so no PC knows the MAC address of another PC yet. Watch what happens:

1. Wait until the PC–switch links are green, then click the **Simulation** tab. Click **Show All/None**, then **Edit Filters**, and tick only **ARP** and **ICMP**.
2. Choose **Add Simple PDU** (the closed envelope) and click PC1, then PC2. Press **Capture then Forward** again and again. Click an envelope to see its layers.

An ICMP envelope appears at PC1 first, but it cannot leave yet. The frames then move in this order:

1. **ARP request** (broadcast). PC1 knows PC2's IP address but not its MAC address, so it asks every device: "Who has 192.168.10.11?" The frame goes to `FFFF.FFFF.FFFF`, the broadcast MAC address. S1 sends it out of all its other ports (this is called *flooding*).
2. **ARP reply** (unicast). Only PC2 answers, directly to PC1.
3. **ICMP Echo Request** (type 8) from PC1 to PC2, then **ICMP Echo Reply** (type 0) back to PC1.

Click **Reset Simulation** and send the PDU again. Now only ICMP events appear, because each PC keeps the other's MAC address in a table called the *ARP cache* (ARP: Lecture 5; ICMP: Lecture 8).

Return to **Realtime** and type `ping 192.168.10.11` in PC1's **Command Prompt**. Four replies arrive, although S1 has no configuration: a switch forwards frames with its factory settings.

L1.3.4 Step 4: Open the router CLI and learn the modes

Cisco routers and switches run **IOS** (Internetwork Operating System). You configure IOS by typing commands in its **command-line interface** (CLI). The CLI has several **modes**, like floors of a building. Each mode accepts different commands, and the prompt shows where you are.

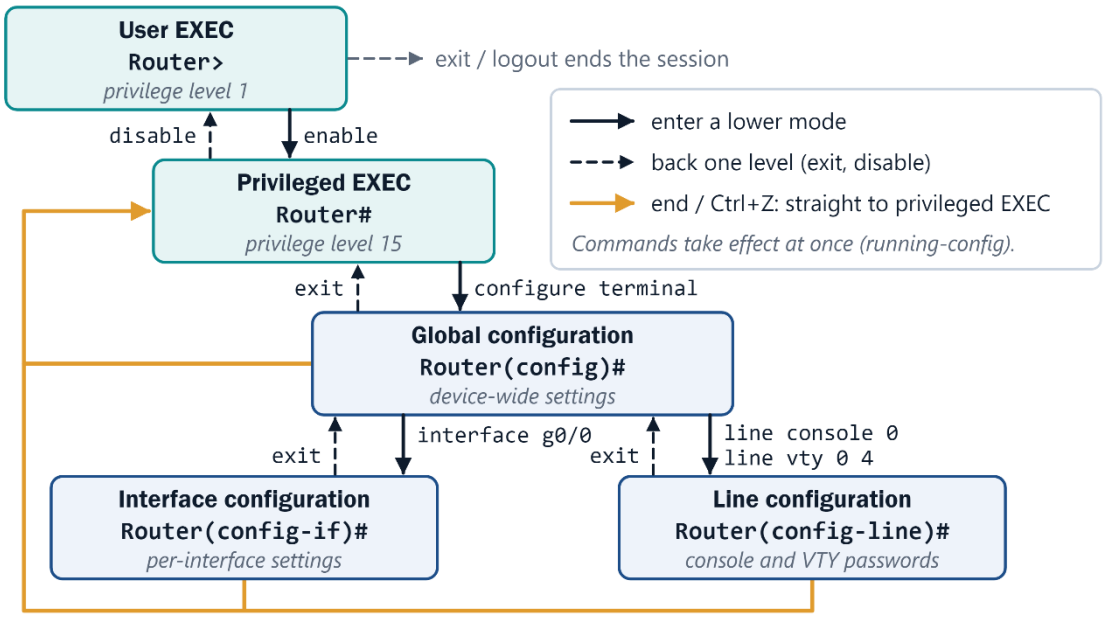


Figure L1.3 IOS command modes. Solid arrows enter a mode, dashed arrows go back one level, and the amber arrow (end or Ctrl+Z) returns straight to privileged EXEC.

Read the figure from the top. `Router>` (user EXEC) only lets you look and test, e.g. `ping`. `Router#` (privileged EXEC) lets you see everything and save. `Router(config)#` (global configuration) changes the whole device. From there, `interface` or `line` opens a sub-mode for one interface (`config-if`) or one access line (`config-line`). `exit` goes back one level; `end` (or Ctrl+Z) jumps straight back to `Router#`.

On PC1 open **Desktop > Terminal**, keep the default settings (9600 bps, 8 data bits, no parity, 1 stop bit, no flow control) and click **OK**. You are now on R1's console line. R1 has no saved configuration, so IOS offers a setup dialog: type `no`, then press Enter until `Router>` appears. Then type the bold commands one at a time (not the grey comments).

```
Router>configure terminal                  ! wrong mode on purpose
       ^
% Invalid input detected at '^' marker.
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#end                            ! two levels up, to Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
```

The first command fails because `configure terminal` does not exist in user EXEC mode: always read the prompt first. To type less, use `?` (lists the allowed commands), Tab (completes a word) and short forms such as `conf t` and `int g0/0`.

L1.3.5 Step 5: Basic configuration of R1

Every new Cisco device gets the same basic settings: a name, passwords, a warning banner and interface addresses. First name R1 and protect access to it:

```
Router#configure terminal
Router(config)#hostname R1                    ! the prompt changes at once
R1(config)#no ip domain-lookup                ! no name lookup after a typo
R1(config)#enable secret class                ! password for privileged EXEC
R1(config)#banner motd #Authorized access only!#
R1(config)#line console 0                     ! the console port
R1(config-line)#password cisco
R1(config-line)#login                         ! ask for the password
R1(config-line)#exit
R1(config)#line vty 0 4                       ! five remote (Telnet) lines
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#exit
R1(config)#service password-encryption        ! hide the line passwords
```

- `no ip domain-lookup`: without it, IOS takes a mistyped word at an EXEC prompt as a host name and waits several seconds trying to find it.
- `enable secret` protects privileged EXEC; IOS stores it as a hash (type 5). Under a line, `password` sets the line password and `login` makes IOS ask for it.
- `banner motd` (message of the day) shows the text between the `#` signs before login: a legal warning, never "Welcome".
- `service password-encryption` hides the line passwords as type 7. Type 7 is easy to reverse, so it only stops casual reading (real security: Lab 10).

Then address the two interfaces, switch them on and save:

```
R1(config)#interface gigabitEthernet 0/0
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#no shutdown                      ! switch the interface on
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
R1(config-if)#interface serial 0/0/0
```

```
R1(config-if)#ip address 10.0.0.1 255.255.255.252
R1(config-if)#clock rate 64000                ! DCE end only: 64 kbps
R1(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down  ! R2 not ready yet
R1(config-if)#end
R1#copy running-config startup-config
Destination filename [startup-config]?                ! press Enter
Building configuration...
[OK]
```

KEY DEFINITION: RUNNING-CONFIG AND STARTUP-CONFIG

Every command works at once, in the **running-config**. It lives in RAM and is lost when the power goes off. `copy running-config startup-config` saves it in NVRAM as the **startup-config**, which the router loads at the next boot. Save after every change.

L1.3.6 Step 6: Configure R2 and S1

Click R2 and open its **CLI** tab; it works like a console connection. Answer `no` to the dialog, type `enable` and repeat the first command block of Step 5 with `hostname R2`. Then address its interfaces. R2 is the DTE end, so it gets no clock rate.

```
R2(config)#interface gigabitEthernet 0/0
R2(config-if)#ip address 192.168.20.1 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#interface serial 0/0/0
R2(config-if)#ip address 10.0.0.2 255.255.255.252
R2(config-if)#no shutdown                     ! now the serial link comes up
R2(config-if)#end
R2#copy running-config startup-config
```

A switch forwards frames without any IP address. It needs one only so that you can manage it over the network. The address goes on a virtual interface, `Vlan1`, called a **switch virtual interface** (SVI). On S1's CLI tab, type `enable` and repeat the first command block of Step 5 with `hostname S1` and `line vty 0 15` (a 2960 has 16 remote lines). Then:

```
S1(config)#interface vlan 1
S1(config-if)#ip address 192.168.10.2 255.255.255.0
S1(config-if)#no shutdown
S1(config-if)#exit
S1(config)#ip default-gateway 192.168.10.1          ! for S1's own traffic
S1(config)#end
S1#copy running-config startup-config
```

L1.4 Verification

First check the interfaces. `show ip interface brief` gives one line per interface:

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.1    YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/0/0            10.0.0.1        YES manual up                    up
Serial0/0/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

Status reports Layer 1 and whether the interface is switched on; **Protocol** reports Layer 2. Both used interfaces must be **up/up**. (PT's 1941 also lists a `Vlan1` line; ignore it.) Other pairs point to the problem:

- **administratively down/down**: switched off by `shutdown`, the router default; type `no shutdown`.
- **down/down**: a Layer 1 problem, such as no cable, a wrong cable, or a far end that is off.
- **up/down**: the cable works but Layer 2 does not, e.g. no clock on a serial link.

Next, `show running-config` lists the active configuration. Check the hostname, banner and addresses, and see that the passwords now appear as `enable secret 5 …` and `password 7 …`, not as plain text.

Finally, test the connections. An IOS ping sends five echoes and prints `!` for a reply and `.` for no reply within 2 s. A router's first ping to an Ethernet host often shows `.!!!!`, because the first echo is lost while ARP finds the MAC address. Record what you observe in each test below.

Table L1.2 Verification tests.

No.	From	Command	Expected result	Why
1	PC1	`ping 192.168.10.1`	Success	The gateway is in LAN1.
2	PC1	`ping 192.168.10.2`	Success	The S1 SVI is up.
3	PC3	`ping 192.168.20.1`	Success	Cross-over link and R2 `G0/0` are up.
4	R1	`ping 10.0.0.2`	`!!!!!`	Serial link up; it needs no ARP.
5	PC1	`telnet 192.168.10.1`	Banner, then `Password:`	The VTY lines have `password` and `login`.
6	PC1	`ping 192.168.20.10`	Destination host unreachable	R1 has no route to LAN2.

Test 6 fails on purpose. R1 knows only its *directly connected* networks (marked **C** in `show ip route`), so it answers with an ICMP "Destination host unreachable" message. Lab 4 adds the missing routes.

L1.5 Common problems

Table L1.3 Common problems. Check from Layer 1 upward: cables, interface status, addresses, then ping hop by hop.

Symptom	Cause	Fix
R2–PC3 link stays red	Straight-through cable	Use a cross-over cable.
Interface shows `administratively down`	`no shutdown` was forgotten	Type `no shutdown` on it.
PC1 reaches R1 but not S1 (192.168.10.2)	S1 `Vlan1` is still shut down	`no shutdown` under `interface vlan 1`.
`% Invalid input detected at '^' marker.`	Typing error or wrong mode	Read the prompt; enter the right mode.
PC2 reaches PC1 but not 10.0.0.1	Wrong default gateway on PC2	Set the gateway to 192.168.10.1.

Symptom	Cause	Fix
Telnet answers "Password required, but none set"	`login` without `password` on the VTY lines	Add `password cisco` under `line vty 0 4`.

L1.6 Discussion questions

1. Name the cable for each link: PC–switch, switch–switch, PC–router (Ethernet), router–router (serial), PC to router console. Why does a PC–router Ethernet link need a cross-over cable?
2. What does the DCE end of a serial link do? How did you make R1 the DCE end in PT?
3. At `R1(config-if)#`, where does `exit` take you, and where does `end`? In which mode do you type `hostname R1`, `password cisco`, `no shutdown` and `copy running-config startup-config`?
4. Where are the running-config and the startup-config stored? You configure R2, forget to save, and then switch R2 off and on. What does R2 show when it restarts, and why?
5. In Simulation mode, why did ARP come before ICMP? Which ARP message was broadcast?
6. How does IOS store the `enable secret` and the encrypted line passwords? Which form is weak?

L1.7 Lab report

- The saved file `LAB01_<StudentID>.pkt`.
- The output of `show ip interface brief` (R1, R2, S1) and `show running-config` (R1).
- The verification table with an extra "Observed" column.
- A labelled screenshot of the Simulation-mode Event List (ARP, then ICMP).
- Your answers to the discussion questions.